Knot Theory

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1 Introduction

First we shall give definitions for knots and lengths and a few examples about different types of lengths and knots to give people a sense of what they look like.

To study knots and the links we shall not only have the rigorous definition. We should use the knot diagram. But the usage of not diagrams brings a problem: if there are two representation diagrams, are they representing the same knot? How should we distinguish between two knots from their diagrams? Reidemeister raised a criterion to tell different diagrams apart. That is Reidemeister move. It has three basic movements: 1. Stretching or straightening 2. Crossing 3. Pull two strings apart and move a string from one side of the crossing to the other side

This is in the following theorem, which is raised in 1933.

Theorem 1.1: Reidemeister move

(1) Any (tame) link (or node) L has a Diagram representation $D(L)$.

(2) $L_1 \sim L_2 \Leftrightarrow D(L_1) \overset{\mathrm{R}}{\sim} D(L_2)$ equivalent up to Reidemeister move.

Definition 1.1: Oriented Link Writhe

Definition 1.2: Geometric Braid

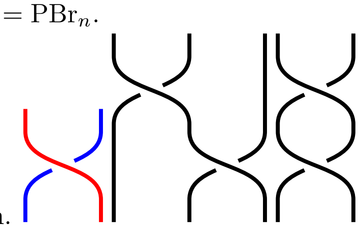
Definition 1.3: Braid group

Consider the space $\mathbb{R}^n - \bigcup_{\tau \text{transposition}} H_\tau$. The symmetric group S_n action on it freely and properly. S_n also acts freely and properly on the space $\mathbb{C}^n - \bigcup_{\tau \text{transposition}} H_\tau \otimes_{\mathbb{R}} \mathbb{C}$. This is called the configuration space. If the order of the space is removed, that is, we only consider each element as the set instead of an ordered product. The configuration space quotient by the symmetric group action, which is called the unconfiguration space.

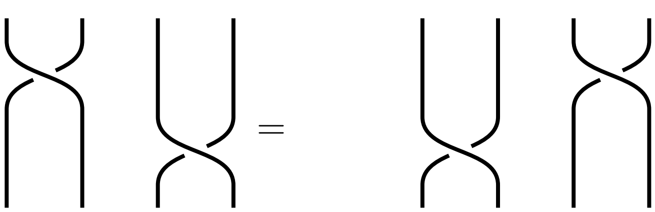
Definition 1.4: Configuration space

The configuration space with order n, denoted Conf_n, is $\mathrm{Conf}_n = \mathbb{C}^n - \bigcup_{\tau \text{transposition}} H_\tau$.
The un-configuration space with order n, denoted UConf_n, is $\mathrm{Conf}_n = \mathbb{C}^n - \bigcup_{\tau \text{transposition}} H_\tau \big/ _{S_n}$.

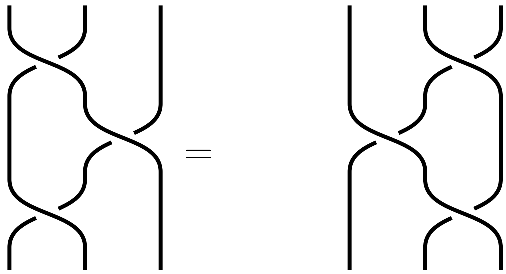
There are some very nice properties of the braid groups. The fundamental group of the configuration space of order n is the braid group, namely $\pi_1(\mathrm{Conf}_n) = \mathrm{Br}_n$. And the fundamental group of the configuration space of order n is the pure braid group, i.e. $\pi_1(\mathrm{UConf}_n) = \mathrm{PBr}_n$.

Then, let's try to assign a knot an orientation. 

Far Commutativity: $\sigma_1\sigma_3 = \sigma_3\sigma_1$



Yang-Baxter Equation: $\sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2$



Braid groups are special cases of Artin groups. Artin group provides a uniform representation of all kinds of groups like the braid groups, the Coxeter groups via an integer-weighted graph, which presents the 'commutator relations' between two elements.

1.0.1 Coxeter groups

Coxeter groups and the dihedral groups